

# Luigi Roberti

## Curriculum Vitæ

Leibniz Universität Hannover  
Institut für Angewandte Mathematik  
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### Personal Information

Place of birth   Brescia (BS), Italy  
Citizenship    Italian  
Gender          Male  
Title          Dr.rer.nat.

### Current Employment

01/10/2025 – [Institut für Angewandte Mathematik, Leibniz Universität Hannover](#),  
Present      Germany.  
Wissenschaftlicher Mitarbeiter – Postdoc (Postdoctoral Research Assistant).

### Past Employment

01/10/2021 – [Faculty of Mathematics, University of Vienna](#), Austria.  
31/08/2025    Universitätsassistent Praedoc (Predoctoral University Assistant).  
01/10/2020 – [Faculty of Mathematics, University of Vienna](#), Austria.  
30/09/2021    Wissenschaftlicher Projektmitarbeiter (Scientific Project Assistant).  
Employment within the project “[Equatorial wave-current interactions](#)”,  
supported by the WWTF grant MA16-009.  
Principal Investigator: Prof. Dr. [ADRIAN CONSTANTIN](#).

### Education

10/2020 – **Dr.rer.nat. in Mathematics**, University of Vienna, Austria.  
08/2025    Thesis: *Mathematical aspects of geophysical fluid flows*.  
Advisor: Prof. Dr. [ADRIAN CONSTANTIN](#).  
10/2018 – **MSc in Mathematics**, University of Vienna, Austria.  
09/2020    Thesis: *On the decrease of velocity with depth in periodic water waves*.  
Advisor: Prof. Dr. [ADRIAN CONSTANTIN](#).  
10/2015 – **BSc in Mathematics**, University of Vienna, Austria.  
08/2018    Thesis: *Hardy's inequality*.  
Advisor: Prof. Dr. [ROLAND DONNINGER](#).  
10/2014 – **BSc in Physics**, University of Vienna, Austria.  
03/2018    Thesis: *Bestimmung der Volumenverteilung eines Aerosols aus der Messung von dessen Extinktionskoeffizienten*.  
Advisor: Prof. Dr. [HELMUTH HORVATH†](#).  
09/2009 – **Diploma di Maturità Scientifica (Science High School Diploma)**,  
07/2014    Liceo Scientifico Statale Nicolò Copernico, Brescia, Italy.

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## Publications

### Preprints

- [1] (With [C. PUNTINI](#)) A note on the lower boundary condition for Ekman flows. [arXiv:2607.29261](#), 11 pp.

### Accepted articles

- [2] (With [C. PUNTINI](#) and [E. STEFANESCU](#)) On large-scale wind-drift ocean currents: An asymptotic approach in spherical coordinates. [arXiv:2602.06473](#), 42 pp., to appear in *J. Fluid Mech.*

### Published articles

- [3] (With [Q. DING](#)) Variational formulation for a model describing the Antarctic Circumpolar Current with variable density. *Differ. Integral Equ.* **39**(7/8) (2026), 497–510. [doi:10.57262/die039-0708-497](#).
- [4] (With [E. STEFANESCU](#)) Global-in-time existence, uniqueness, and stability of solutions to a model of the Antarctic Circumpolar Current. *Discrete Contin. Dyn. Syst.* **49** (2026), 289–310. [doi:10.3934/dcds.2025169](#).
- [5] (With [B.-V. MATIOC](#) and [CH. WALKER](#)) Quasilinear parabolic equations with superlinear nonlinearities in critical spaces. *J. Differ. Equ.* **429** (2025), 283–317. [doi:10.1016/j.jde.2025.02.039](#).
- [6] (With [Q. DING](#)) Stratified ocean gyres with Stuart-type vortices. *Ann. Mat. Pura Appl.* **203** (2024), 2847–2862. [doi:10.1007/s10231-024-01469-5](#).
- [7] (With [B.-V. MATIOC](#)) Weak and classical solutions to an asymptotic model for atmospheric flows. *J. Differ. Equ.* **367** (2023), 603–624. [doi:10.1016/j.jde.2023.05.023](#).
- [8] The surface current of Ekman flows with time-dependent eddy viscosity. *Comm. Pure Appl. Anal.* **21**(7) (2022), 2463–2477. [doi:10.3934/cpaa.2022064](#).
- [9] The Ekman spiral for piecewise-constant eddy viscosity. *Appl. Anal.* **101**(15) (2022), 5528–5536. [doi:10.1080/00036811.2021.1896709](#).
- [10] Perturbation analysis for the surface deflection angle of Ekman-type flows with variable eddy viscosity. *J. Math. Fluid Mech.* **23**(3) (2021), No. 57. [doi:10.1007/s00021-021-00586-y](#).
- [11] On the decrease of velocity with depth in irrotational periodic water waves. *Monatsh. Math.* **193**(3) (2020), 671–682. [doi:10.1007/s00605-020-01451-2](#).

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## Conferences and Seminars

### Talks

- 2026 Invited talk at workshop [Analysis and Partial Differential Equations 2026](#), Leibniz Universität Hannover, Germany.
- Invited talk at [the 15<sup>th</sup> AIMS Conference](#), Athens, Greece.
- Talk at [SIAM Conference on Nonlinear Waves and Coherent Structures \(NWCS26\)](#), Montréal, Canada.
- Invited talk in working seminar, Hangzhou Normal University, China.

- Invited talk at workshop Fluid Dynamics and Spectral Theory of Differential Operators, Nanjing University, China.
- 2025 Invited talk at [Bielefeld Analysis Seminar](#), Universität Bielefeld, Germany.  
Talk at [Oberseminar Analysis und Theoretische Physik](#), Leibniz Universität Hannover, Germany.  
Invited talk at workshop [Modelling of Fluid Propagation: Mathematical Theory and Numerical Approximation](#), International Centre for Mathematical Meetings (CIEM), Castro Urdiales, Spain.  
Invited talk at [Conference on Mathematics of Wave Phenomena](#), Karlsruhe Institute of Technology, Germany.
- 2024 Invited talk at [Mathematical Sciences Seminar](#), University College Cork, Ireland.  
Talk at workshop [Mathematical Theory of Water Waves](#), Lund University, Sweden.  
Talk at [PDE Afternoon Seminar](#), University of Vienna, Austria.  
Invited talk at [MCMP Seminar](#), University of Vienna, Austria.
- 2023 Talk at [ÖMG Tagung 2023](#), TU Graz, Austria.
- 2022 Talk at [DMV-Jahrestagung 2022](#), Freie Universität Berlin, Germany.  
Talk at [Seminar Applied and Computational PDE](#), University of Vienna, Austria.
- 2020 Talk at [Seminar: Applied PDE](#), University of Vienna, Austria.
- 2019 Talk at [Complex Analysis Seminar](#), University of Vienna, Austria.

### **Additional conferences and programmes attended**

- 2025 Workshop [Mathematical Advances in Geophysical Fluid Dynamics](#), Mathematisches Forschungsinstitut Oberwolfach, Germany.
- 2024 [EWM-EMS Summer School: Water Waves and Nonlinear Dispersive Equations](#), Mittag-Leffler-Institut, Sweden.  
[Fluid Flows – Analysis and Modelling. Conference in Honour of Robin S. Johnson’s 80th Birthday](#), University of Vienna, Austria.
- 2023 [Aspects of Nonlinear Evolution. Conference in Honour of Joachim Escher’s 60th Birthday](#), Leibniz Universität Hannover, Germany.  
[Workshop on Nonlinear Dispersive Waves](#) (online attendance), University College Cork, Ireland.
- 2021 [Summer School of the Vienna School of Mathematics](#), Weißensee, Austria.

## **Teaching**

### **Leibniz Universität Hannover**

- 2026W [Übung zu Numerische Mathematik I](#) (Exercise class to Numerical Mathematics I, in German).  
[Übung zu Mathematik I für Life Sciences und Geowissenschaften](#) (Exercise class to Mathematics I for Life Sciences and Geoscience, in German).  
[Seminar Leray-Hopf Solutions to the Navier-Stokes Equations](#) (together with Prof. Dr. JOACHIM ESCHER, in English).

- 2026S [Funktionentheorie für das Lehramt](#) (Lecture on complex analysis for prospective teachers, in German).  
[Übung zu Funktionentheorie für das Lehramt](#) (Exercise class to my lecture on complex analysis for prospective teachers, in German).

### University of Vienna

- 2024W [Proseminar “Komplexe und Harmonische Analysis”](#) (Exercise class “Complex and Harmonic Analysis”, in German).  
 2024S [Proseminar “Analysis 2”](#) (Exercise class “Analysis 2”, in German).  
 2023W [Proseminar “Analysis und Lineare Algebra 1”](#) (Exercise class “Analysis and Linear Algebra 1”, in German).  
 2023S [Übungen zu “Funktionalanalysis”](#) (Exercise class to “Functional Analysis”, in German).  
 2022W [Übungen zu “Partielle Differentialgleichungen”](#) (Exercise class to “Partial Differential Equations”, in German).  
 2022S [Übungen zur Einführung in das mathematische Arbeiten und Rechenübungen](#) (Exercise class to Introduction to mathematics and calculations, in German).

### (Co-)Supervision

- 2025 BSc thesis of Stefano Mazzeo, University of Vienna (co-supervised with Dr. [JÖRG WEBER](#)).

### Peer Review Activity

Referee for *Appl. Math. Comput.* / *Appl. Math. Lett.* / *Comm. Pure Appl. Anal.* / *J. Differ. Equ.* / *J. Math. Phys.* / *J. Phys. Oceanogr.* / *Monats. Math.* / *Nonlinear Anal. Real World Appl.* / *Pure Appl. Geophys.* / *Qual. Theory Dyn. Syst.*

### Training

- 2025 Workshop [Diversity in Practice](#), University of Vienna.

### Computer Skills and Competence

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|-----------------------|-------------------------------------------------------------------------------------------------------------------|
| Operating systems     | Working knowledge of Windows and Linux OS.                                                                        |
| Programming languages | Working knowledge of Python, Fortran, LaTeX language and some of its text editors (Overleaf, Texmaker, TeXworks). |
| Softwares             | Working knowledge of Matlab, Wolfram Mathematica, Gnuplot.                                                        |

### Platforms

-  [arXiv](#)  
 [Google Scholar](#)  
 [ORCID](#)  
 [ResearchGate](#)

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## Languages

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|---------|----------------------------------------------------------------------------|
| Italian | Native speaker.                                                            |
| English | Excellent knowledge and understanding of both written and spoken language. |
| German  | Excellent knowledge and understanding of both written and spoken language. |

Hannover, September 29, 2026